COMPSCI 682 Assignment 2

Download the starter code (assignment2.zip) from the Canvas assignment page.

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Setup

Please familiarize yourself with the recommended workflow in the *Working remotely on Google Colaboratory* section of the **Software Setup** page before starting the assignment.

The assignment would be using Python = 3.12

Once you have completed all Colab notebooks, proceed to the instructions in Submitting your work below.

Colab: If using colab, make a new folder, (ex. `cs682`) in google drive and upload the `assignment2` folder. After opening any notebook in colab you will need to uncomment and run the first code cell. This will mount google drive and cd into the `assignment2` directory. Do not forget to answer the inline questions. To write your response click on the text and then click the edit pencil icon that appears in the top right.

Download data: Once you have the starter code, you will need to download the CIFAR-10 dataset.

Inside the colab notebook, there is already a cell where you can run the following:

```
%cd ./cs682/datasets
!bash get_datasets.sh
%cd ../../
```

Alternatively, on your local comptuer run the following from the `assignment2` directory:

```
cd ./cs682/datasets
bash get_datasets.sh
```

Compile the Cython extension: Convolutional Neural Networks require a very efficient implementation. We have implemented the functionality using Cython; you will need to compile the Cython extension before you can run the code.

Inside the colab notebook, there is already a cell where you can run the following:

```
%cd cs682
!python setup.py build_ext --inplace
%cd ..
```

Alternatively, on your local comptuer run the following from the `assignment2/cs682` directory:

```
python setup.py build_ext --inplace
```

NOTE: Check this page if you are using windows and having the "unable to find vcvarsall.bat" error.

Jupyter Notebook: After you have the CIFAR-10 data, if you are not using colab you should start the Jupyter Notebook server from the `assignment2` directory. If you are unfamiliar with Jupyter, you should read our Jupyter tutorial.

NOTE: If you are working in a virtual environment on OSX, you may encounter errors with matplotlib due to the issues described here. You can work around this issue by starting the Jupyter server using the `start_jupyter_osx.sh` script from the `assignment2` directory; the script assumes that your virtual environment is named `.env`.

Goals

In this assignment you will practice writing backpropagation code, and training Neural Networks and Convolutional Neural Networks. The goals of this assignment are as follows:

- understand **Neural Networks** and how they are arranged in layered architectures
- understand and be able to implement (vectorized) **backpropagation**
- implement various **update rules** used to optimize Neural Networks
- implement **batch normalization** for training deep networks
- implement **dropout** to regularize networks
- effectively **cross-validate** and find the best hyperparameters for a Neural Network architecture
- understand the architecture of **Convolutional Neural Networks** and gain experience with training these models on data

Q1: Fully-connected Neural Network (16 points)

The notebook **FullyConnectedNets.ipynb** will introduce you to our modular layer design, and then use those layers to implement fully-connected networks of arbitrary depth. To optimize these models you will implement several popular update rules.

Q2: Batch Normalization (34 points)

In the notebook **BatchNormalization.ipynb** you will implement batch normalization, and use it to train deep fully-connected networks.

Q3: Dropout (10 points)

The notebook **Dropout.ipynb** will help you implement Dropout and explore its effects on model generalization.

Q4: ConvNet on CIFAR-10 (30 points)

In the notebook **ConvolutionalNetworks.ipynb** you will implement several new layers that are commonly used in convolutional networks. You will train a (shallow) convolutional network on CIFAR-10, and it will then be up to you to train the best network that you can.

Q5: PyTorch (10 points)

For this last part, you will be working in PyTorch, a popular deep learning framework.

Open up **PyTorch.ipynb**. There, you will learn how the framework works, culminating in training a convolutional network of your own design on CIFAR-10 to get the best performance you can.

Submitting your work

Important. Please make sure that the submitted notebooks have been run and the cell outputs are visible.

Once you have completed all notebooks and filled out the necessary code, you need to follow the below instructions to submit your work:

To make sure everything is working properly, **remember to do a clean run ("Kernel -> Restart & Run All") after you finish work for each notebook** and submit the final version with all the outputs.

1. Generate a zip file of your code (`.py` and `.ipynb`) called `<UmassID>.zip` (For email address `arnaik@umass.edu` - zip file name is `arnaik.zip`). Please ensure you do not include the dataset folder in the zip.

On your local machine (macOS, Linux, or WSL on Windows), run the provided script from the `assignment2/` directory, e.g. `bash collectSubmission.sh arnaik`, to produce `<UmassID>.zip` (if you run it in a terminal without your UMass ID, it asks for it). In Colab, run `!bash collectSubmission.sh <UmassID>` in a code cell of any notebook.

2. Convert all notebooks (`.ipynb` files) into a single PDF file. In colab this can be done by selecting `File -> Print -> Print to PDF`.

3. Please submit `<UmassID>.zip` and the PDF to Gradescope.